

COMPLETE SETUP GUIDE

Real-Time Adaptive Noise Cancellation (MATLAB + Laptop Mic + Mobile Mic)

PHASE 1 — SYSTEM & SOFTWARE REQUIREMENTS

1 Hardware required

- Laptop with **built-in microphone**
- Smartphone (Android or iPhone)
- USB data cable
- **Headphones / earphones** (mandatory)

 Do **not** use laptop speakers. Feedback will break ANC.

2 MATLAB requirements

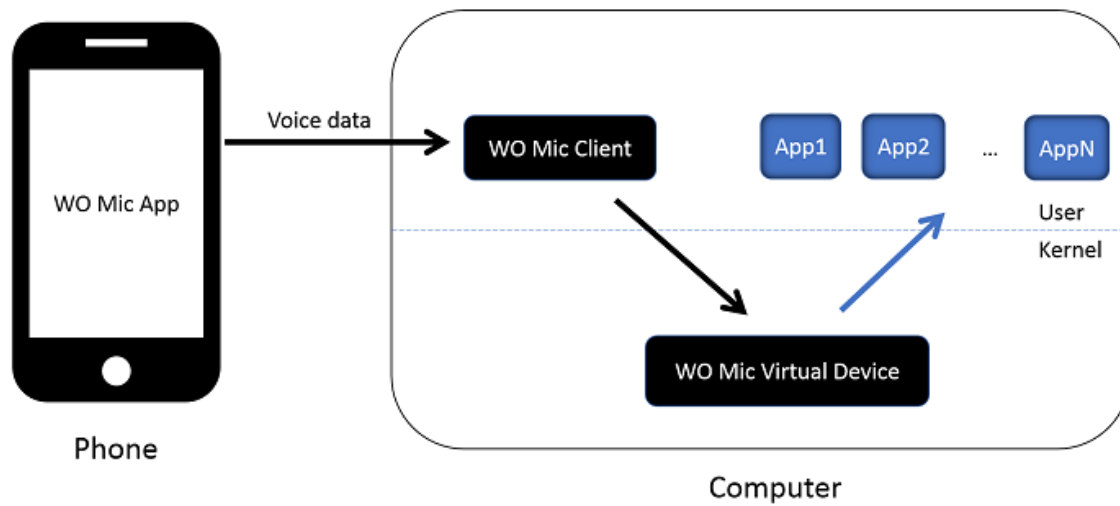
- MATLAB **R2020a or newer** (you used R2023a ✓)
- Required toolboxes:
 - **Audio Toolbox**
 - **DSP System Toolbox**

Verify in MATLAB:

ver

If either toolbox is missing → stop and install.

PHASE 2 — SET UP PHONE AS MICROPHONE (WO MIC)



Connect ✕

Transport type

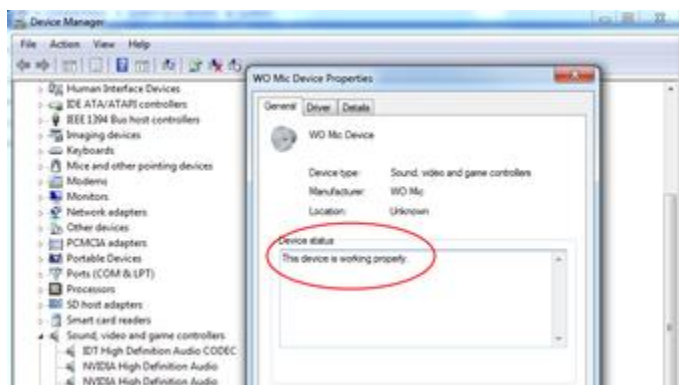
- ☒ Wi-Fi
- ☐ Bluetooth
- ☐ USB
- ☐ Wi-Fi Direct

Details

Phone IP address:

Tips:

1. Connect phone and PC to same Wi-Fi AP (hot spot).
2. Select Wi-Fi as transport on phone and start the server. It shows phone IP address at panel top.
3. Enter that IP address into above edit box.



3 Install on PHONE

- Install **WO Mic** from Play Store / App Store
 - Open once, then close
-

4 Install on LAPTOP (Windows)

- Install **WO Mic Client**
 - Install **WO Mic Virtual Audio Driver**
 - **Restart laptop** (mandatory)
-

5 USB connection (preferred)

Android only:

1. Settings → About phone → tap *Build number* 7 times
2. Developer options → enable **USB debugging**

Then:

- Connect phone to laptop via USB
 - Allow **USB debugging** popup
 - USB mode → **File Transfer (MTP)**
-

6 Start WO Mic

Phone:

- Open WO Mic
- Transport → **USB**
- Audio source → Microphone
- Tap **START**

Laptop:

- Open WO Mic Client

- Connection → **USB**
 - Click **Connect**
 - Status must show **Connected**
-

7 Verify in Windows (DO NOT SKIP)

- Sound settings → Input
- Select **WO Mic Device**
- Speak near phone → input bar must move

If this fails → MATLAB will NOT work.

PHASE 3 — VERIFY IN MATLAB

8 Detect audio devices

audiodevinfo

Then list names:

```
devices = audiodevinfo;
```

```
for k = 1:length(devices.input)
```

```
    fprintf('Input %d: %s\n', k, devices.input(k).Name);
```

```
end
```

You must see:

- **Microphone Array (Realtek...)** → laptop mic
 - **Microphone (WO Mic Device)** → phone mic
-

PHASE 4 — SIGNAL VALIDATION (CRITICAL)

9 Test laptop mic alone

```
frameLength = 256;
primaryMic = audioDeviceReader( ...
    'Device', 'Microphone Array (Realtek(R) Audio)', ...
    'SamplesPerFrame', frameLength);
x = primaryMic();
release(primaryMic);
mean(abs(x))
```

Expected: 0.01 – 0.5

10 Test phone mic alone

```
refMic = audioDeviceReader( ...
    'Device', 'Microphone (WO Mic Device)', ...
    'SamplesPerFrame', frameLength);
n = refMic();
release(refMic);
mean(abs(n))
```

Expected: 0.01 – 0.5

1 1 Test both together (buffer priming)

```
primaryMic = audioDeviceReader( ...
    'Device', 'Microphone Array (Realtek(R) Audio)', ...
    'SamplesPerFrame', frameLength);

refMic = audioDeviceReader( ...
    'Device', 'Microphone (WO Mic Device)', ...
    'SamplesPerFrame', frameLength);
```

```
for k = 1:20
```

```
    x = primaryMic();
```

```
    n = refMic();
```

```
end
```

```
fprintf('Primary: %.4f | Reference: %.4f\n', ...
```

```
    mean(abs(x)), mean(abs(n)));
```

```
release(primaryMic);
```

```
release(refMic);
```

Both values must be **non-zero**.

NOW U ARE ALL SETUP , JUST RUN THE MAIN CODE